

Dante (sockd) dynamic-routing templates

Revision: 1 **Last modified:** 2026-07-01T00:00:00Z **Status:** DESIGN / templates only — NOT yet wired into a running stack.

This directory holds the **template** the future control-plane config-compiler (Stream B) renders to add dynamic-mode per-tunnel SOCKS egress to Dante. It is **additive**: the rendered route blocks are **appended after** the shipped `config/dante/sockd.conf` content (preserved verbatim) to build the deployed config — the base config (internal: `:34080`, external: `proxy-dante`, `socksmethod: none`, and its client pass / socks pass rules) is never modified or removed (§11.4.122).

Why concatenation, not a native include: Dante v1.4.4 has **no include directive** (CONFIRMED — `sockd -V` rejects both `include: "..."` and `include "..."` with a syntax error: `qa-results/p4_templates/dante_parse.txt`, TEST 1 & 3). Squid supports `include` (so the Squid template *is* included); Dante does not, so the compiler concatenates instead. The end result is the same additive guarantee: base content untouched, route blocks added.

Files

File	Purpose
<code>dynamic-routing.sockd.conf.tmpl</code>	Per-tunnel route <code>{ ... }</code> blocks directing matching destinations out through that tunnel's <code>gluetun</code> egress.

Placeholders (the compiler fills these)

Placeholder	Meaning
<code>{{TUNNEL_NAME}}</code>	Per-tunnel label (comment only), e.g. <code>tun_<profile></code> .
<code>{{TARGET_CIDR}}</code>	Destination(s) routed through this tunnel (CIDR or host).
<code>{{PEER_HOST}}</code>	

Placeholder	Meaning
	gluetun container host exposing this tunnel's egress proxy.
{{PEER_PORT}}	That gluetun proxy port (SOCKS upstream for chaining).

The compiler renders **one route block per tunnel/target**.

How the compiler fills + wires them

1. Read VPN profiles + target routes from PostgreSQL.
2. Substitute the placeholders, one route block per tunnel/target.
3. **Concatenate**: write the shipped config/dante/sockd.conf content verbatim, then append the rendered route blocks, into the deployed config (Dante has no include directive — see note above). The base lines stay byte-for-byte.
4. Apply via **SIGHUP** to the sockd mother process — only on a **structural** route change (new/removed tunnel or target).

proxyprotocol / egress topology

The template chains matching destinations through the tunnel's gluetun proxy so bytes leave via that VPN netns. Set proxyprotocol to match what gluetun exposes:

- proxyprotocol: socks_v5 — if a SOCKS upstream is exposed (template default).
- proxyprotocol: http_connect — for gluetun's HTTP proxy (:8888).

Alternative interface-bind topology (one container = one netns, §3.2): instead of chaining, bind egress to the tunnel interface —

```
route {
    from: 0.0.0.0/0 to: {{TARGET_CIDR}}
    via: {{TUNNEL_IFACE}}
}
```

The compiler selects the form per deployment; the shipped template ships the proxy-chaining form as the portable default.

Confirmed facts & honest gaps (§11.4.6)

- **FACT (F_spikes_G1-G4.md §G3)**: SIGHUP reloads sockd config and does **not** drop an active ESTABLISHED SOCKS session (20/20 chunks across a mid-stream reload, curl exit 0). Adding/removing a route is live-session safe.

- **UNCONFIRMED:** whether an in-flight session whose **route changed** keeps its OLD path or is re-evaluated — owed to **P9** (concurrent / repeated / route-change live captured-evidence test).
- **FACT (qa-results/p4_templates/dante_parse.txt, TEST 2):** the rendered route { ... } block **verifies clean** — sockd -V -f <conf> exit **0** (Dante v1.4.4, vimagick/dante:latest, via rootless podman run --rm). Only an unrelated "uid 0 not recommended" warning (running -V as root), not a route syntax issue.
- **FACT (TEST 1 & 3):** sockd -V **REJECTS** an include line (both include: and bare include) with a syntax error — hence the concatenation model above.
- **UNCONFIRMED:** full live SOCKS egress through a real per-tunnel upstream + the route-change-mid-session behaviour — owed to **P9** against the real stack (sockd -V validates syntax + interface, not live proxy chaining). Recorded honestly, not faked.

Validation

See qa-results/p4_templates/ (gitignored) for the rendered example and any captured Dante parse attempt + its honest result.